POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Microprocessor and Computer Architecture | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How is a computer organized? Explain Von Neumann Architecture and Harvard Architecture with necessary diagrams. 7

   b) Draw the block diagram of an 8085 microprocessor and explain, in brief, all the functional units in it. 8

2. a) How the instructions are classified in 8085 microprocessor? Explain with examples 8

   b) What is the need for timing diagram Draw the timing diagram of instruction ***MOV A,B*** 7

3. a) Write the features of 8086 microprocessor. Explain the addressing modes in 8086 Microprocessor. 7

   b) Design a very simple CPU, with its state diagram, RTL codes, register section and ALU, for the given set of instructions. The system uses 6-bit Address register, 6-bit Program counter, 8-bit Data register and 2-bitInstruction register. 8

Instruction	Instruction Code	Operation
ADD	00AAAAAA	AC←AC + M[AAAAAA]
OR	01AAAAAA	AC←AC v M[AAAAAA]
JMP	10AAAAAA	GOTO AAAAAA
INC	11XXXXXX	AC←AC+2

4. a) Design hardwired control unit for the CPU described in question 3b. Also, briefly explain its operation, how the control signals are generated and execution performed. 7

   b) Explain the design of a micro-sequencer control unit with vertical microcode for the CPU described in question 3b. 8

5. a) Define RTL. Mention the RTL codes for shift-add algorithm and trace the multiplication of 13 and 11. Also, verify the obtained product. 8

   OR

   How signed numbers are represented? Write RTL codes for Booth's algorithm and use it to multiply -13 and 3.Also, verify the obtained product.

   b) Differentiate between Segmentation and Paging? Describe the two most common replacement algorithms related to design issues in Cache memory. 7

6. a) What do you understand by DMA? Explain it with suitable diagram. 8

   OR

   Describe with the proper diagram the working principle of I/O processors.

   b) What are the different topologies used to interconnect MIMD computers? Describe with suitable diagrams. 7

7. Write short notes on: (**Any two**) 2×5

   a) A 3×3 Wallace Tree

   b) Multicore processor architecture

   c) Microprocessor Vs Micro controller